Quantum Money

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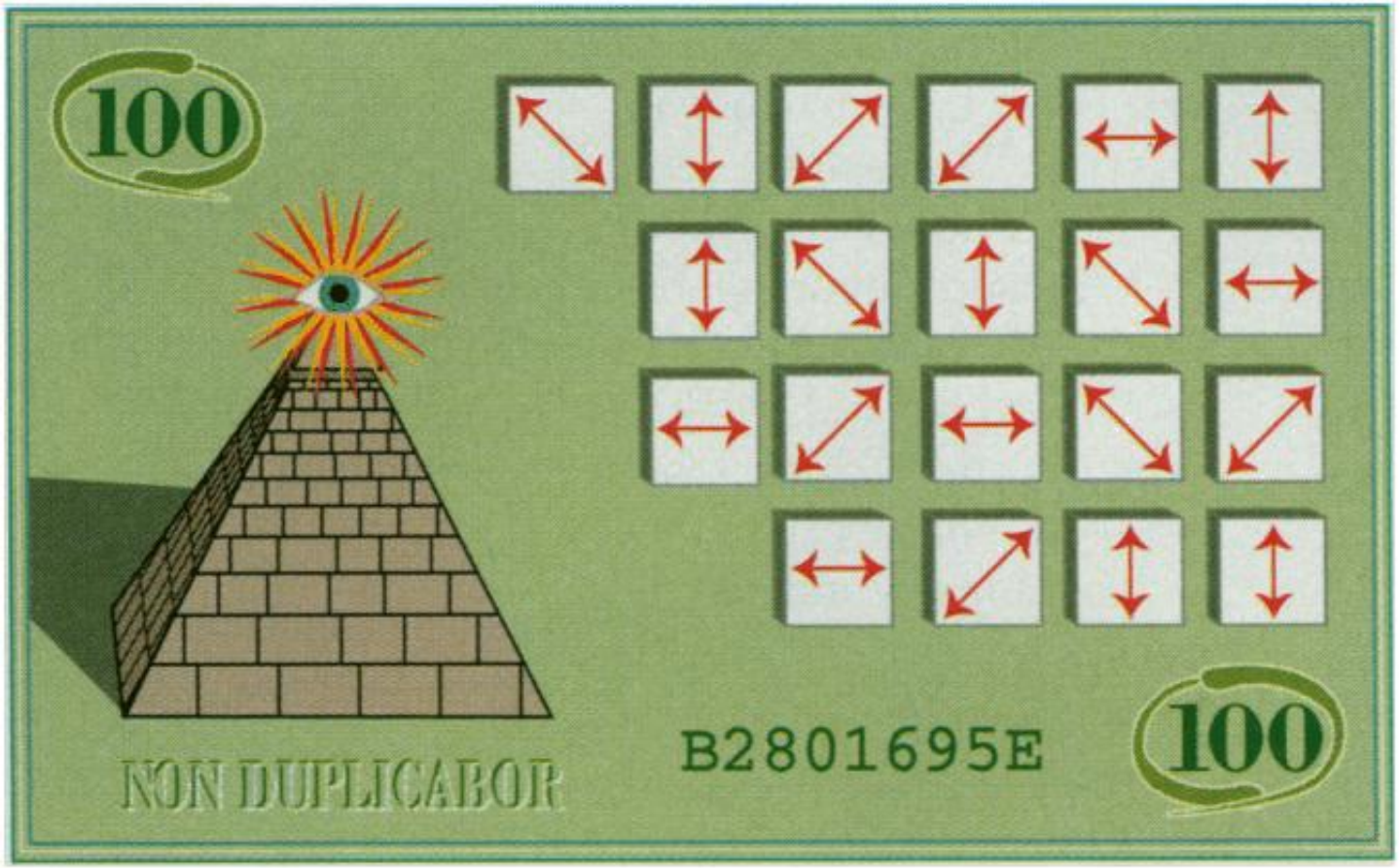
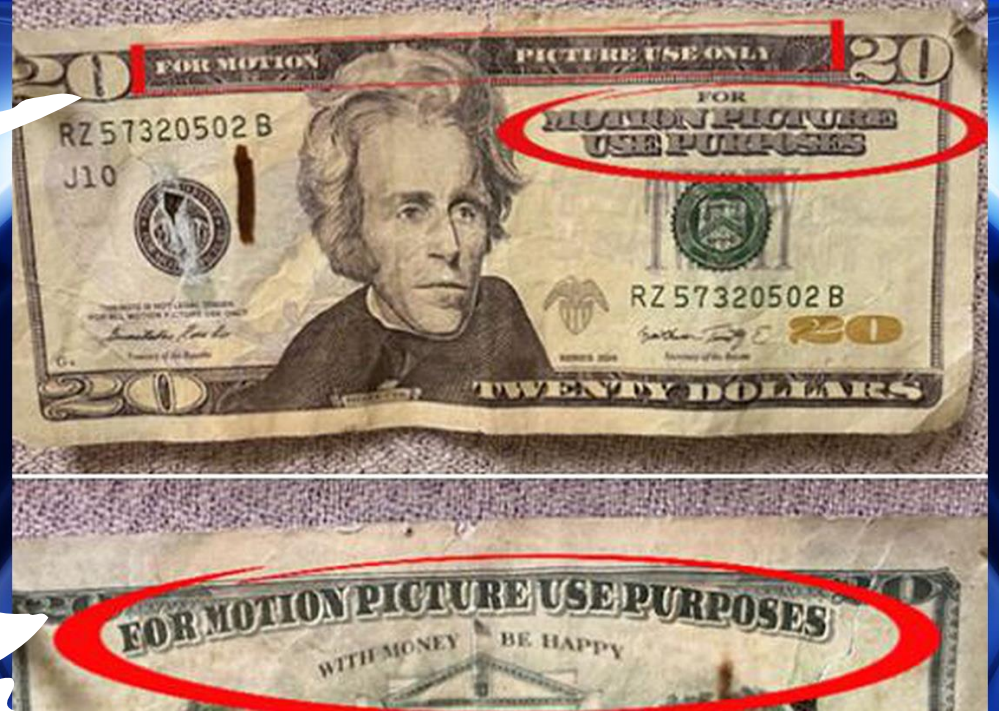


Based on an article by Aaronson et al. https://cacm.acm.org/research/quantum-money/

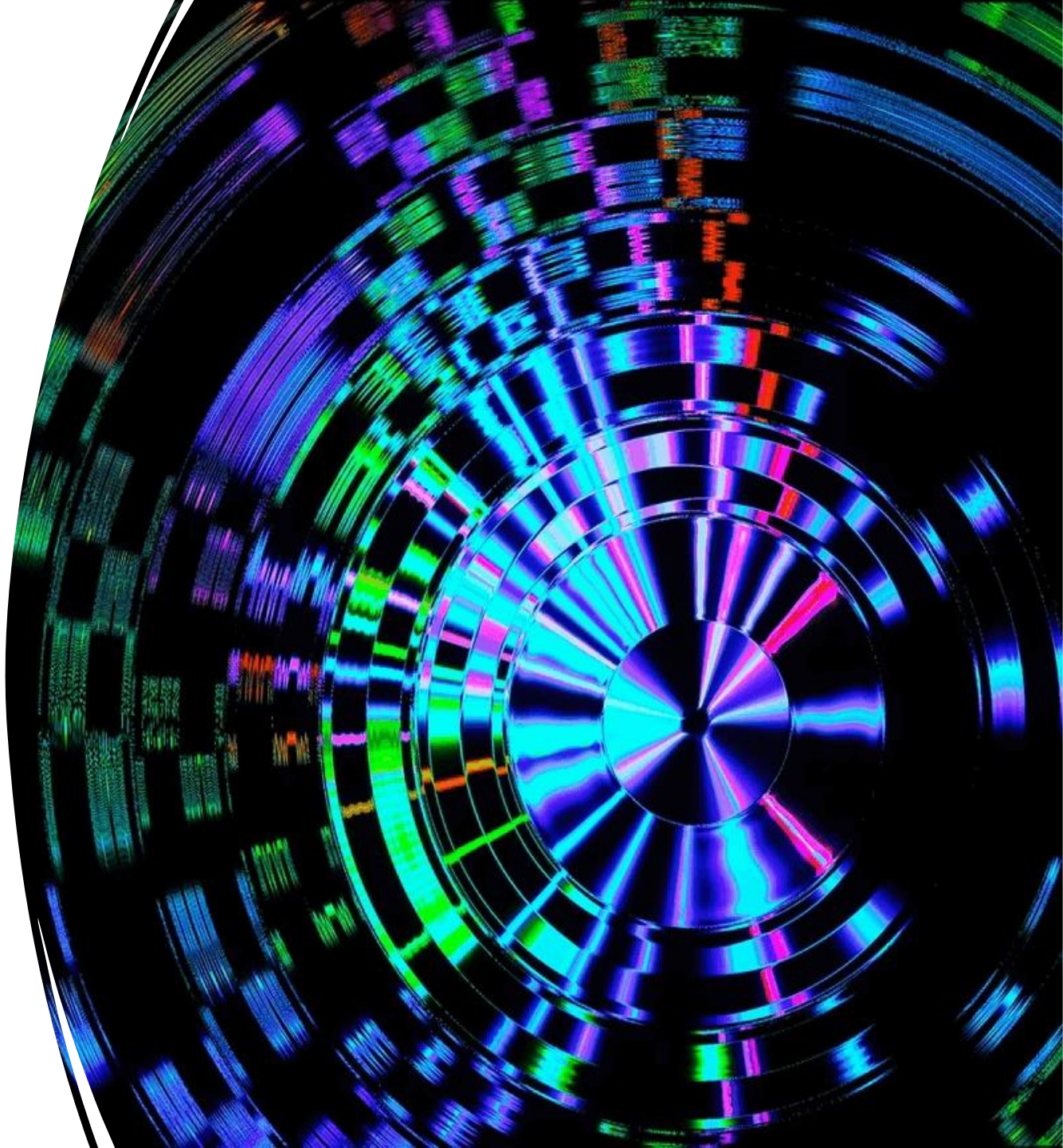
Classical Money

Money Counterfeit



Introduction

Quantum Money is a theoretical form of currency that uses quantum states to secure transactions. It is based on the principle that **quantum states cannot be perfectly duplicated**, making it impossible to forge quantum money by including quantum systems in its design.



Quantum Preliminaries

- Qubits

$$|0\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \qquad |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

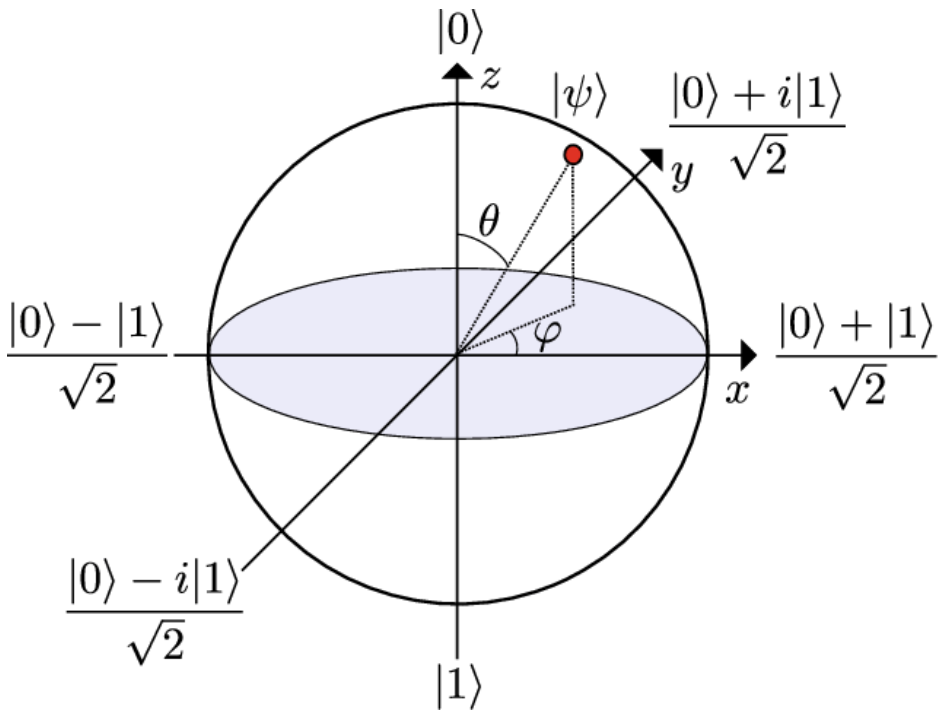
- Superpositions

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}} \quad |-\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$$

- Measurement changes state

$$|0\rangle \text{ with probability } |\alpha|^2$$
$$|1\rangle \text{ with probability } |\beta|^2$$
$$|\alpha|^2 + |\beta|^2 = 1$$



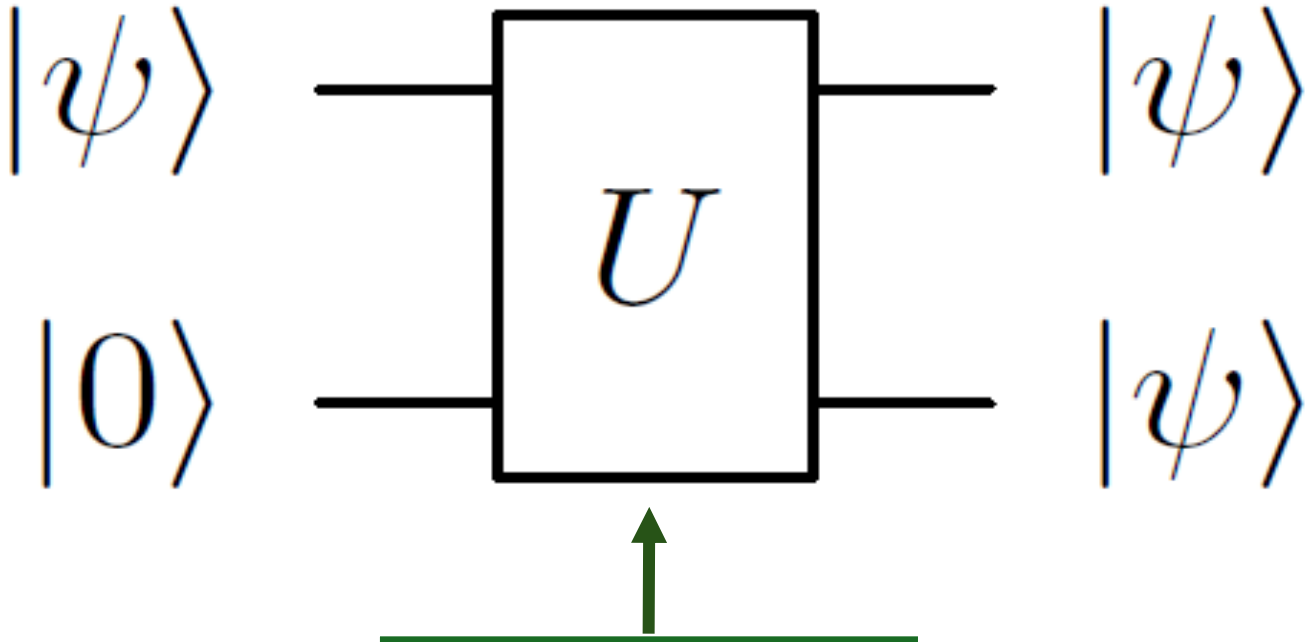
- Can measure in different bases

$$|0\rangle = \frac{|+\rangle+|-\rangle}{\sqrt{2}} \rightarrow \{|0\rangle, |1\rangle\}, \quad \{|+\rangle, |-\rangle\}$$

- Once you measure it, you get the same value

No-Cloning Theorem:

There is **no perfect quantum copy machine** that can copy **any** quantum state you feed it.



This is not unitary!

Quantum Money

Private Key

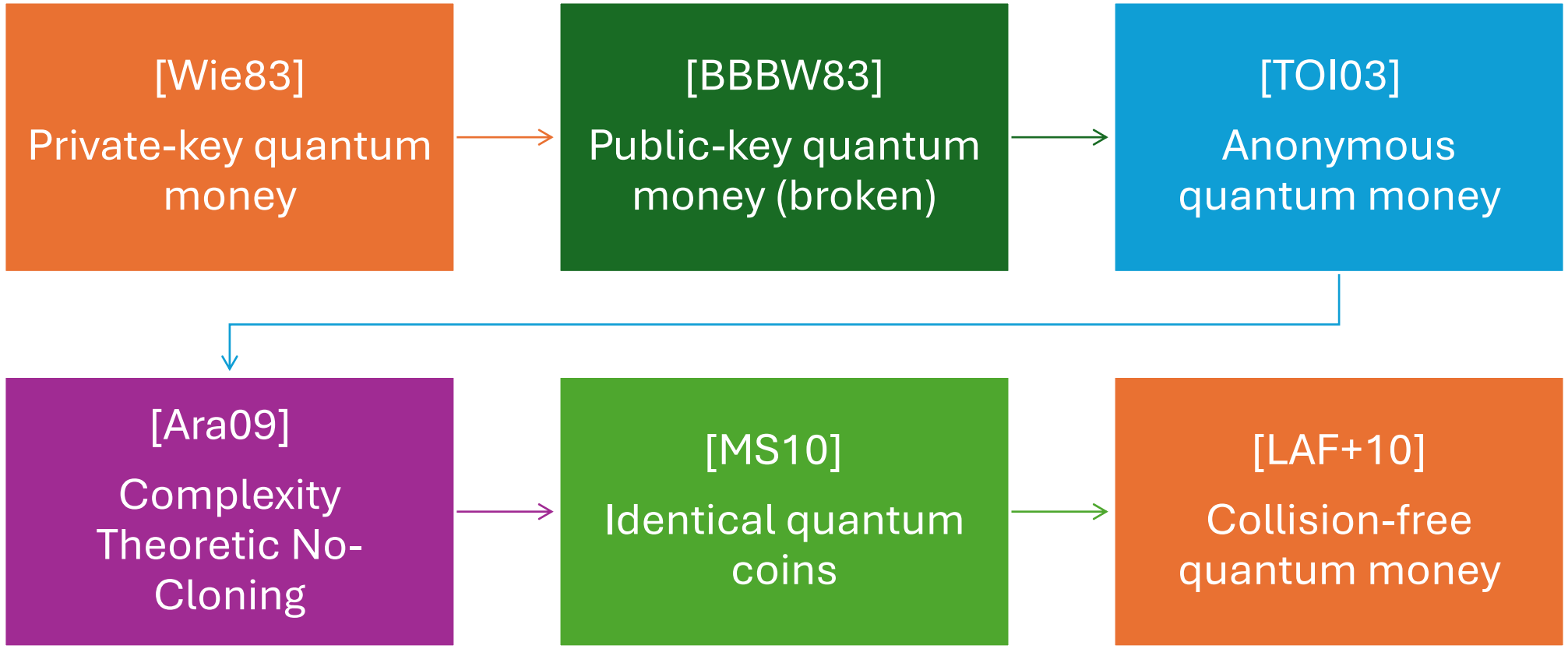
- Mint produces bills consisting of qubits.
- Anyone can store or trade it.
- **Only the mint can verify validity.**

Public Key

- Mint produces bills consisting of qubits.
- Anyone can store or trade it.
- **Anyone can verify validity, without communicating to the mint.**

Ideal for practical purposes

Timeline

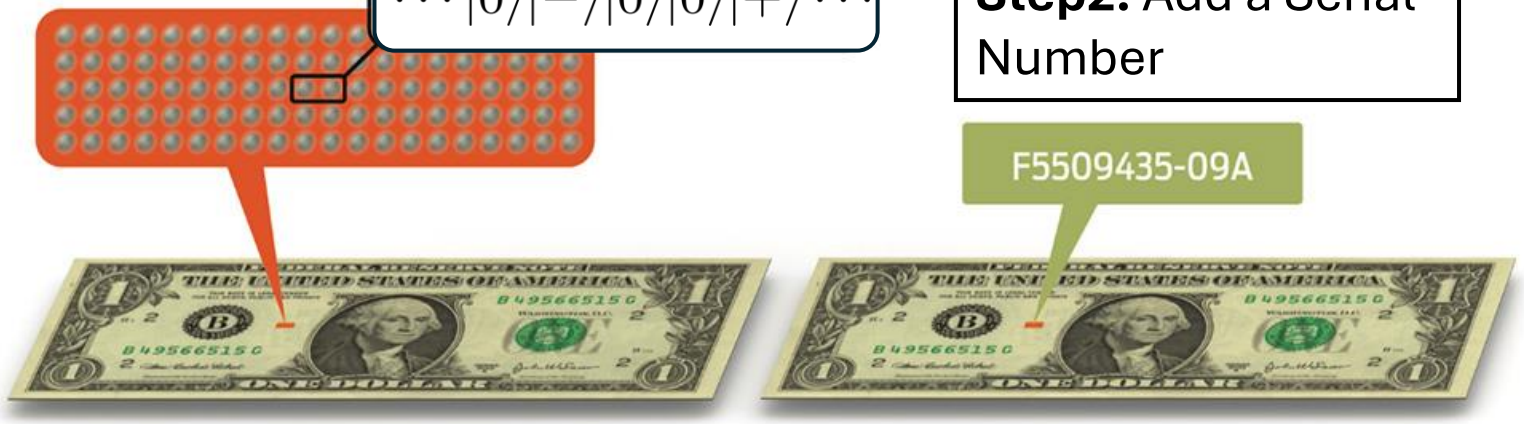

[Wie83]
Private-key quantum money
[BBBW83]
Public-key quantum money (broken)
[TOI03]
Anonymous quantum money
[Ara09]
Complexity Theoretic No-Cloning
[MS10]
Identical quantum coins
[LAF+10]
Collision-free quantum money

Wiesner's Private-Key Quantum Money

Step1: Insert n qubits into bills.

$$\cdots|1\rangle|0\rangle|+\rangle|1\rangle|-\rangle\cdots$$
$$\cdots|-\rangle|+\rangle|0\rangle|1\rangle|-\rangle\cdots$$
$$\cdots|0\rangle|-\rangle|0\rangle|0\rangle|+\rangle\cdots$$

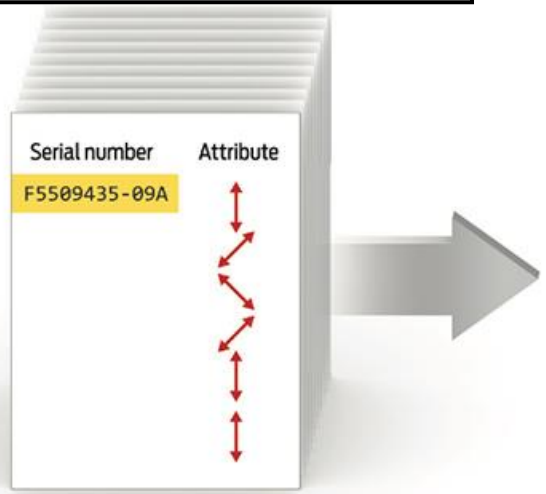
Step2: Add a Serial Number

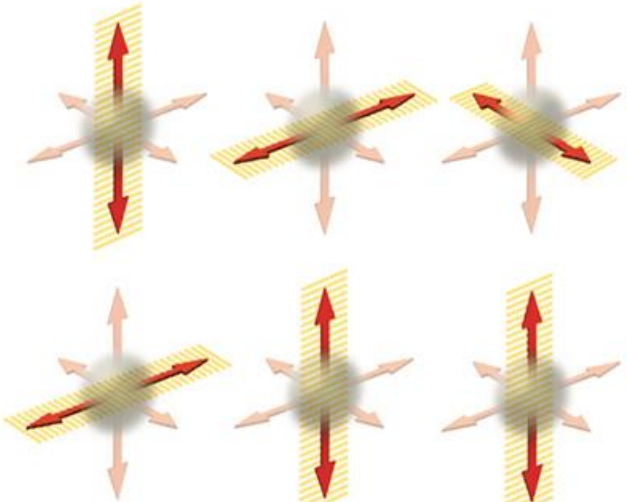


Step3: Keep the s.no. and states in a database

Step4: (Verification) Measure the qubits in the bases according to the database.

Step5: If measurements match, declare validity





Source: Quantum cash and the end of counterfeiting, IEEE Spectrum, May 2012 (Edited)

Public Key Quantum Money

The mint can mint it.

Anyone can verify it without communicating with the mint.

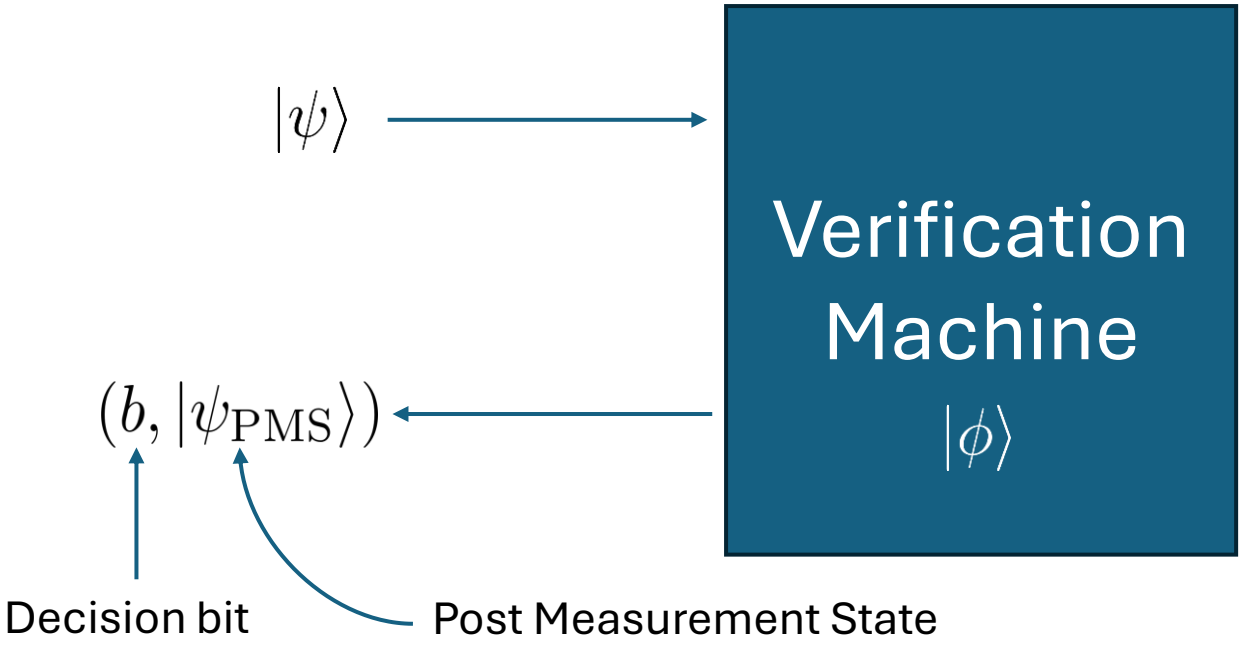
No one (except possibly the mint) can copy it.

Challenge: The verification algorithm can be used by a forger!

Public Key Setting

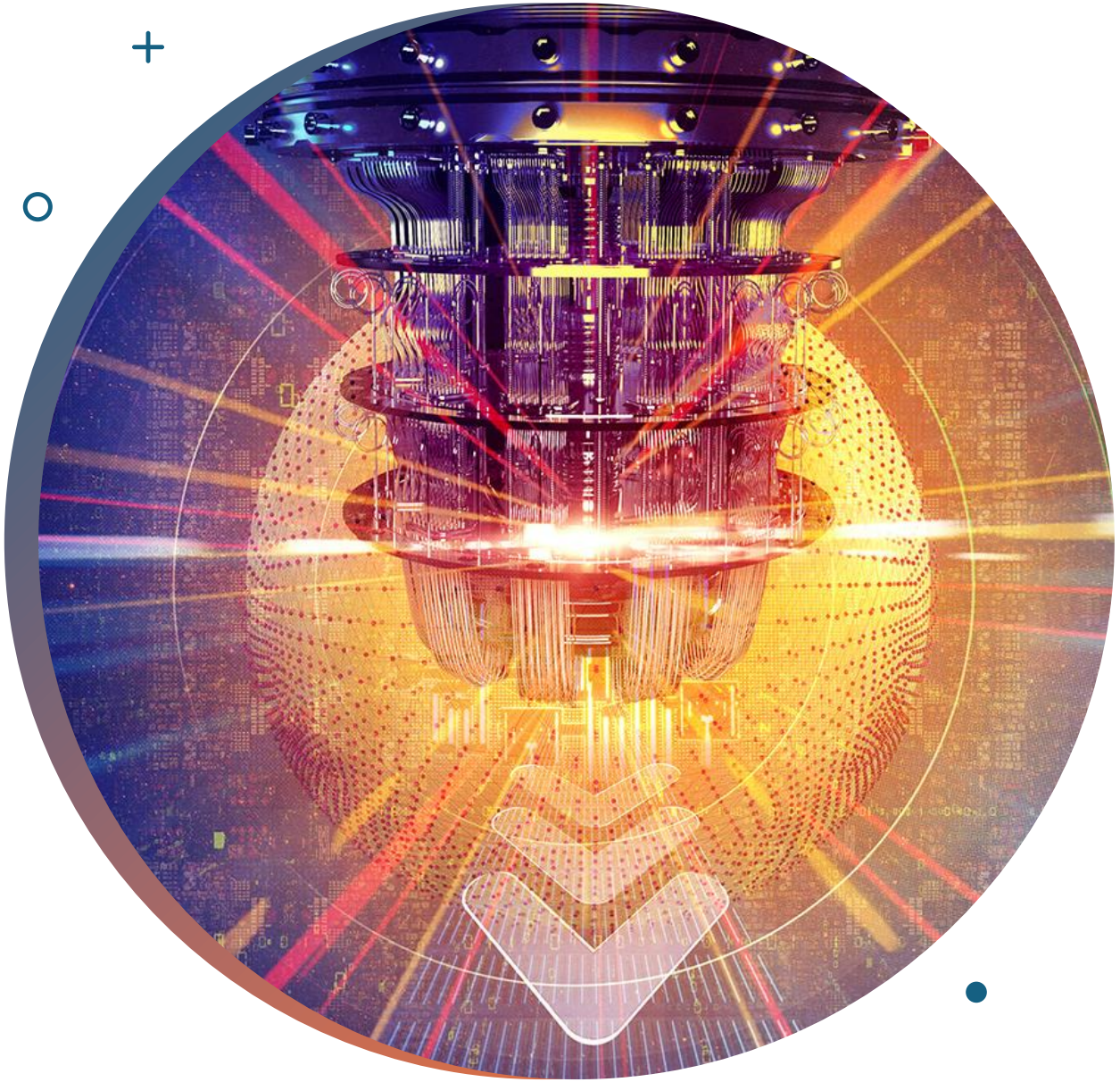
- The standard no-cloning theorem is not good enough. Why?

Complexity Theoretic No-Cloning Theorem:
If the machine is a **black box**, it can fall into the hands of a counterfeiter without compromising the quantum money scheme.



Benefits of Quantum Money

- Enhanced security
- Reduced fraud
- Nearly impossible to counterfeit



Challenges and Limitations

- Technological feasibility
- Regulatory hurdles
- Need for widespread infrastructure
- Physical Realization
- Minor Errors

Conclusion

- Public-key quantum money is one of few quantum protocols that does something that is **truly impossible** classically, even under cryptographic assumptions.
- **Further applications:** Quantum Copy Protection
- However, economic and scalability challenges remain in its implementation.

Thank You!

Any questions?